

Membrane curvature regulates Ups1 dependent phosphatidic acid transfer across lipid bilayers

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eLife Assessment

This **fundamental** study combines in vitro reconstitution experiments and molecular dynamics simulations to elucidate how membrane lipids are transported from the outer to the inner membrane of mitochondria. The authors provide **convincing** evidence that a positive membrane curvature is critical for membrane lipid extraction. The work will be of broad interest to cell biologists and biochemists.

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Abstract

Mitochondria are essential organelles in eukaryotic cells, enclosed by two membranes with distinct compositions and functions. In addition to the endoplasmic reticulum, mitochondria are major sites of cellular lipids production. Cardiolipin, for example, is exclusively synthesized in the mitochondrial inner membrane. This requires the precursor lipid phosphatidic acid to be imported from the endoplasmic reticulum to the mitochondrial outer membrane. Subsequently phosphatidic acid is transferred to the inner membrane by the lipid transfer protein Ups1/PRELID1. The regulation of this process, the role of membrane physico-chemical properties, and the mechanisms by which energy barriers are overcome during lipid extraction and insertion remain poorly understood.

Here, we demonstrate that Ups1 exhibits a strong preference for binding to positively curved membrane regions. Our findings reveal that phosphatidic acid extraction is energetically favored at these membrane domains, leading to enhanced lipid transfer between membranes with high positive curvature. We further show, that lipid extraction is the rate-limiting step in the transfer cycle. Our data suggest that Ups1 membrane binding is modulated by pH, lipid composition, and membrane morphology, pointing to a complex, multipartite regulatory network underlying intra-mitochondrial lipid transfer.

Introduction

Eukaryotic cells exhibit a high degree of compartmentalization characterized by the presence of various membrane-enclosed organelles. On one hand, organellar membranes act as barriers creating specialized reaction spaces. On the other hand, essential biochemical pathways such as lipid biosynthesis and ATP production occur at or within these membranes. Consequently, the composition of organellar membranes is highly diverse in terms of both protein and lipid content and is tailored to their physiological function (van Meer et al., 2008 [↗](#)).

The endoplasmic reticulum (ER) is a major site of lipid biosynthesis (Bishop & Bell, 1988 [↗](#)). From there, lipids are distributed and transferred to various organelles, including mitochondria. Lipid transfer to mitochondria is non-vesicular and is believed to be mediated by lipid transfer proteins at or near membrane contact sites (Flis & Daum, 2013 [↗](#); Osman et al., 2011 [↗](#); Tatsuta et al., 2014 [↗](#)). In mitochondria, some of these lipids are further processed to other lipid species, highlighting the crucial role of mitochondria in cellular lipid biosynthesis. Among these lipids, phosphatidic acid (PA) is converted through a multi-step process in the mitochondrial inner membrane (MIM) to the mitochondrial signature lipid cardiolipin (CL) (Osman et al., 2011 [↗](#)). With its two negative charges and strong cone shape, CL plays a crucial role in shaping the biophysical properties of mitochondrial membranes (Konar et al., 2023 [↗](#); Olofsson & Sparr, 2013 [↗](#); Wilson et al., 2019 [↗](#)). Beyond this structural role, CL is crucial for many mitochondrial processes such as the functionality of the respiratory chain complexes, the ATP synthase and the protein import machinery, as well as for mitochondrial dynamics, apoptosis, and mitophagy (Dudek, 2017 [↗](#); Gonzalez et al., 2008 [↗](#); Muhleip et al., 2019 [↗](#); Paradies et al., 2019 [↗](#); van der Laan et al., 2007 [↗](#)). Consequently, impairments in CL metabolism result in mitochondrial dysfunction and have been linked to various pathologies, including cardiomyopathies, metabolic disorders, neurodegenerative diseases and cancer. Furthermore, CL levels decline with aging, possibly contributing to increased reactive oxygen species production and a heightened susceptibility to apoptosis (Ahmadpour et al., 2020 [↗](#); Falabella et al., 2021 [↗](#); Paradies et al., 2019 [↗](#); Prola & Pilot-Storck, 2022 [↗](#)).

CL biosynthesis in the MIM begins with the precursor lipid PA, which must be imported from the ER and then transported from the mitochondrial outer membrane (MOM) to the MIM. This non-vesicular transport of PA across the intermembrane space (IMS) is mediated by a conserved heterodimeric protein complex found across eukaryotes. In yeast, this complex consists of Ups1 (PRELID1 in mammals), which functions as the lipid-binding protein, and Mdm35 (TRIAP1), which stabilizes Ups1 and protects it from degradation by the i-AAA protease Yme1 and the metallopeptidase Atp23 (Osman et al., 2009 [↗](#); Potting et al., 2013 [↗](#); Potting et al., 2010 [↗](#); Tamura et al., 2009 [↗](#); Tamura et al., 2010 [↗](#)).

Crystal structures of the Ups1/Mdm35 complex have revealed that the lipid-binding pocket of Ups1 is formed by a seven-stranded β -sheet which, together with an α -helix on one side, creates a cavity that accommodates a single PA molecule. The PA headgroup is deeply embedded within the pocket, while its acyl chains extend toward the entrance. The acyl chains are shielded from the aqueous environment by the Ω -loop, which is thought to function as a lid (Lu et al., 2020 [↗](#); Miliara et al., 2015 [↗](#); Watanabe et al., 2015 [↗](#); Yu et al., 2015 [↗](#)). In vitro studies have shown that Ups1 effectively transfers PA between artificial membrane vesicles (Connerth et al., 2012 [↗](#); Miliara et al., 2015 [↗](#); Watanabe et al., 2015 [↗](#)). They also demonstrated that Mdm35 is required to keep Ups1 soluble when not membrane-bound but upon membrane binding of Ups1 dissociates and remains in solution (Connerth et al., 2012 [↗](#)).

These findings support the following model: Ups1 binds to the IMS surface of the MOM, likely dissociating from Mdm35. A PA molecule is extracted by Ups1 from the MOM, and Ups1 then reassembles with Mdm35 to form a transferable complex. PA is subsequently shuttled by Ups1/Mdm35 through the aqueous IMS. Finally, Ups1 binds to the IMS surface of the MIM, dissociates from Mdm35, and releases the PA into the MIM (Connerth et al., 2012 [↗](#); Tamura et al., 2020 [↗](#)).

The molecular details of many of these steps are poorly understood. It is, for example, unclear what gives directionality to the process, how membrane characteristics regulate Ups1 activity, or how the energy barrier is overcome to break lipid-lipid interactions, extract a lipid from one membrane, and insert it into another. To address this, we combined in vitro and in silico experiments to quantitatively analyze the interaction of Ups1 with membranes and investigate how these interactions regulate lipid transfer.

Results

Ups1 exhibits pH-dependent membrane binding with a transition point within the pH range of the IMS

Understanding the membrane-binding properties of Ups1 is crucial for a comprehensive understanding of its lipid transfer mechanism. To analyze the binding characteristics in detail, it is essential to ensure that the interaction occurs via the protein's natural binding site without interference from additional factors. Ups1 is proposed to interact with negatively charged membranes via positively charged residues (Connerth et al., 2012 [↗](#)). In previous studies, constructs containing an N-terminal His-tag were commonly used for purification and biochemical characterization (Connerth et al., 2012 [↗](#); Lu et al., 2020 [↗](#); Watanabe et al., 2015 [↗](#); Yu et al., 2015 [↗](#)). Given its proximity to the putative membrane-binding region, the His-tag might potentially affect membrane binding (Figure 1A [↗](#)). With a pKa of approximately 6.0, it introduces additional positive charges in a pH-dependent manner, potentially masking the intrinsic binding properties of Ups1. To address this issue, we constructed and expressed a cleavable Ups1 variant (Supplementary Figure 1B [↗](#)) to eliminate the potential influence of the His-tag and assessed its membrane interactions.

Flotation assays revealed strong membrane binding of Ups1 at pH 5.5, which markedly decreased at pH 7.5, with a transition point around pH 6.5 (Figure 1B-E [↗](#)). No obvious differences between Ups1 and His-Ups1 could be detected (Figure 1F [↗](#), Supplementary Figure 1C [↗](#)). Reflectometric interference spectroscopy (RIFS) measurements confirmed these findings, yielding a dissociation constant of $K_D = 0.38 \pm 0.05 \mu\text{M}$ at pH 5.5, while no binding was resolvable at pH 7.4 (Figure 1G [↗](#)).

Our findings reveal a pronounced pH-dependent binding of Ups1 to PA-containing membranes, raising the question about the molecular mechanisms. Strikingly, binding decreases as pH increases, despite PA becoming more negatively charged under these conditions. The pKa of PA has been reported to be around 6.9 when in proximity to primary amines, such as those found in PE. Based on this consideration, one would expect stronger binding at higher pH due to enhanced electrostatic interactions, a behavior observed with other PA-binding proteins (Young et al., 2010 [↗](#); Zegarinska et al., 2018 [↗](#); Zhou et al., 2024 [↗](#)). However, Ups1 exhibits the opposite trend, indicating that additional factors influence its pH-dependent membrane interactions. Our experiments excluded the involvement of histidines from the His-tag, shifting attention toward intrinsic features of the protein.

pKa predictions indicate that histidines at positions 5, 9, 33, and 100 may have values within the critical pH range, depending on the conformation of Ups1. It is conceivable that protonation events at these sites could induce conformational changes underlying the observed pH-dependent

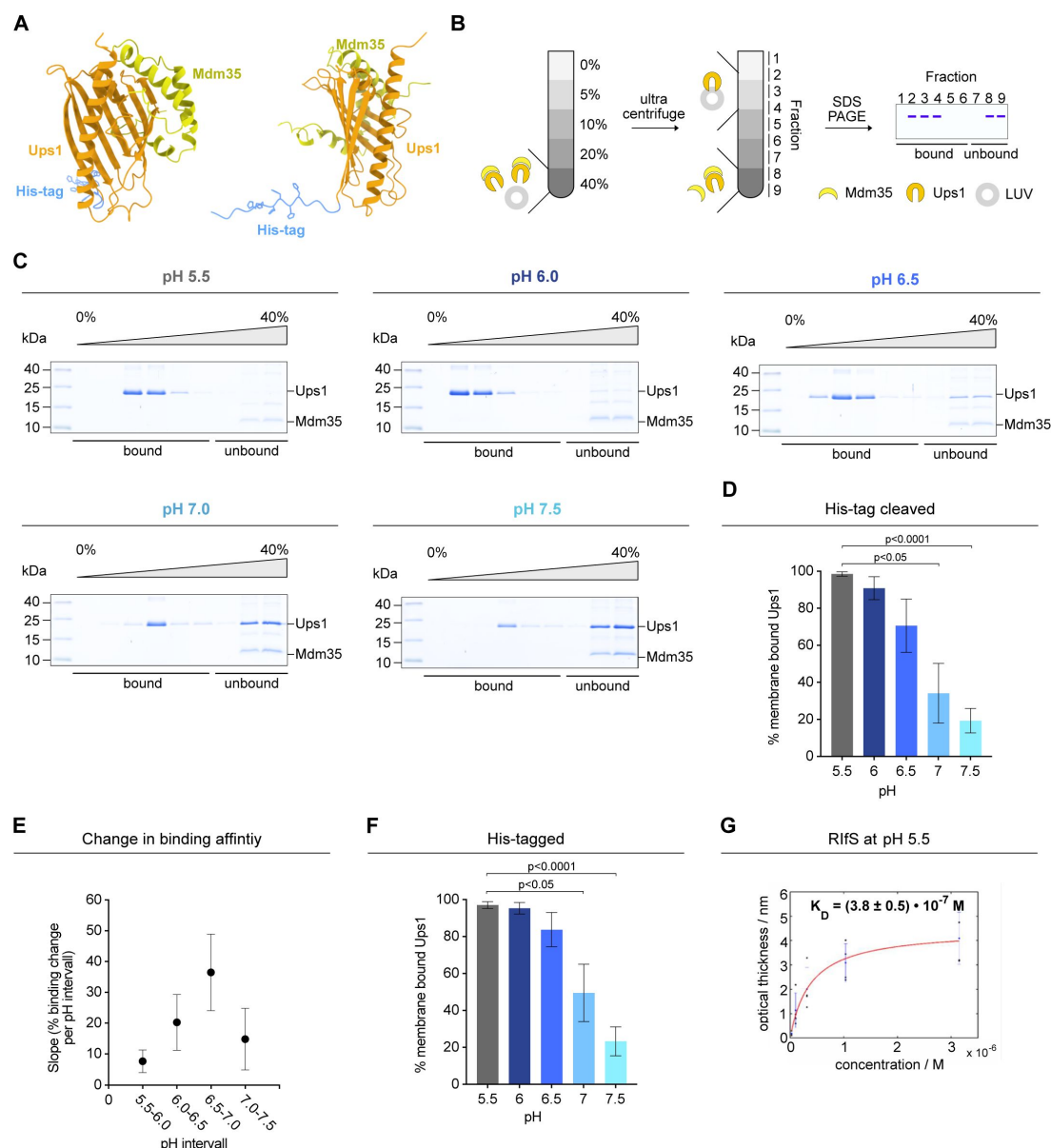


Figure 1.

pH-dependent binding of Ups1 to membranes reveals a shift in affinity across the physiological range of the IMS

(A) Ribbon diagram of His-tagged Ups1 in complex with Mdm35 as predicted by AlphaFold 3 (Abramson et al., 2024). **(B)** Schematic diagram of the flotation assay used to study protein membrane interactions (details see Methods). **(C)** Flotation analyses using untagged Ups1/Mdm35 and LUVs composed of 50% PC, 30% PA, 19.875% PE, 0.125% TF488-PE at indicated pH values. A representative image from three independent experiments is shown for each condition. **(D)** The percentage of membrane-bound Ups1 relative to total Ups1 quantified from Coomassie-stained gels of (C). Data represent the mean $\pm$ SD from three independent experiments. Significant differences relative to pH 5.5 are indicated. **(E)** Change in % membrane-bound Ups1 across the indicated pH intervals based on (D). Each data point represents the binding change between indicated pH values calculated using linear regression. Data are shown as mean slope $\pm$ SE from three independent experiments. **(F)** Quantification of membrane-bound His1-Ups1/Mdm35 from flotation assays using the same liposomes as in (C) at the indicated pH values. Data represent the mean $\pm$ SD from three independent experiments. Significant differences relative to pH 5.5 are indicated. Representative gels are shown in **Supplementary Figure 1C**. **(G)** Dissociation constant (K_D) of Ups1/Mdm35 binding to membranes was obtained by a Langmuir fit of the optical thickness (red line).

membrane interactions. Additionally, the positioning of Histidine 33 within a positively charged patch previously associated with membrane binding (Lu et al., 2020 [↗](#); Yu et al., 2015 [↗](#)) raises the possibility that its protonation state directly affects electrostatic interactions with the membrane. Clarifying the molecular mechanism underlying the pH-dependent membrane binding of Ups1, as well as its physiological relevance (see discussion), will be an important objective for future investigation.

Ups1 exhibits curvature-dependent membrane binding, preferentially associating with regions of high positive curvature

The interaction of Ups1 with membranes has been shown to depend on electrostatic interactions, yet its dependency on other membrane characteristics like membrane morphology remains unexplored. Mitochondria exhibit a striking ultra structure, with their characteristic cristae architecture generating membranes with complex geometries (Barbot & Meinecke, 2016 [↗](#); Sjostrand, 1953 [↗](#); Zick et al., 2009 [↗](#)). Beyond this large-scale structural organization, local membrane morphology is influenced by factors such as protein integration, membrane undulations and lipid composition (Graham & Kozlov, 2010 [↗](#); Khalifat et al., 2008 [↗](#); Lee et al., 2025 [↗](#); Tarasenko & Meinecke, 2021 [↗](#)). For example, proteins with local curvature-inducing properties have been identified in both the inner and outer mitochondrial membranes, highlighting that local curvature changes are integral features of both compartments (Barbot et al., 2015 [↗](#); Callegari et al., 2025 [↗](#); Davies et al., 2012 [↗](#); Diederichs et al., 2020 [↗](#); Hessenberger et al., 2017 [↗](#); Takeda et al., 2021 [↗](#); Tarasenko et al., 2017 [↗](#)). Conversely, membrane curvature plays a crucial role in specific protein-membrane interactions, influencing not only protein binding but also spatial distribution and functional activity (Antonny, 2011 [↗](#); Komatsu et al., 2000 [↗](#); Larsen et al., 2020 [↗](#); Sugiura et al., 2019 [↗](#); Wang et al., 2023 [↗](#)).

To investigate whether membrane morphology influences Ups1 binding, we first employed an *in silico* approach using coarse-grained molecular dynamics (CG-MD) simulations. Curvature within a lipid bilayer system was induced by applying anisotropic compression along the x-axis, allowing expansion along the z-axis. The system was then equilibrated to enable curvature-induced lipid partitioning. Ups1 was introduced into the simulation, and its partitioning free energy ($\Delta\Delta F$) was determined as a function of membrane curvature (**Figure 2A** [↗](#)).

Across a curvature range from -0.188 nm^{-1} ($K_{\min}$) to $+0.188 \text{ nm}^{-1}$ ($K_{\max}$), we observed a significant change in partitioning free energy of $\Delta\Delta F = 9.2 \pm 1.5 \text{ k}_B\text{T}$. Thus, the binding free energy decreased with increasing positive curvature, indicating a preferential localization of Ups1 to membrane regions of high positive curvature (**Figure 2B** [↗](#)). These *in silico* results suggest that Ups1 has a curvature-sensing capability.

To validate these computational findings, we conducted biochemical assays using liposomes of varying diameters to mimic different degrees of membrane curvature. Liposomes were generated by extrusion through membranes with pore sizes of 50 nm and 400 nm to produce vesicles with high and low positive membrane curvature, respectively. Additionally, 50 nm-extruded liposomes were sonicated, resulting in a population that includes vesicles with even smaller diameters. Experiments were performed at pH 7.0, which is within the physiological pH range of the mitochondrial IMS, ensuring that the observed interactions reflect the physiological binding characteristics. Following this experimental setup, liposome co-flotation assays were performed to assess curvature-dependent binding. A relatively weak binding of 31% was observed for 400 nm extruded liposomes, while binding increased to 42% for 50 nm extruded liposomes and reached 70% for sonicated liposomes (**Figure 2C&D** [↗](#)). In agreement with the flotation analyses, Ups1 preferentially bound to small vesicles in co-sedimentation assays (**Figure 2I** [↗](#)) with a gradual increase in association as membrane curvature increased (**Figure 2E&F** [↗](#)). This shows a pronounced and statistically significant curvature-dependent binding pattern, with Ups1 preferentially associating with smaller, highly curved liposomes. Together, these experimental

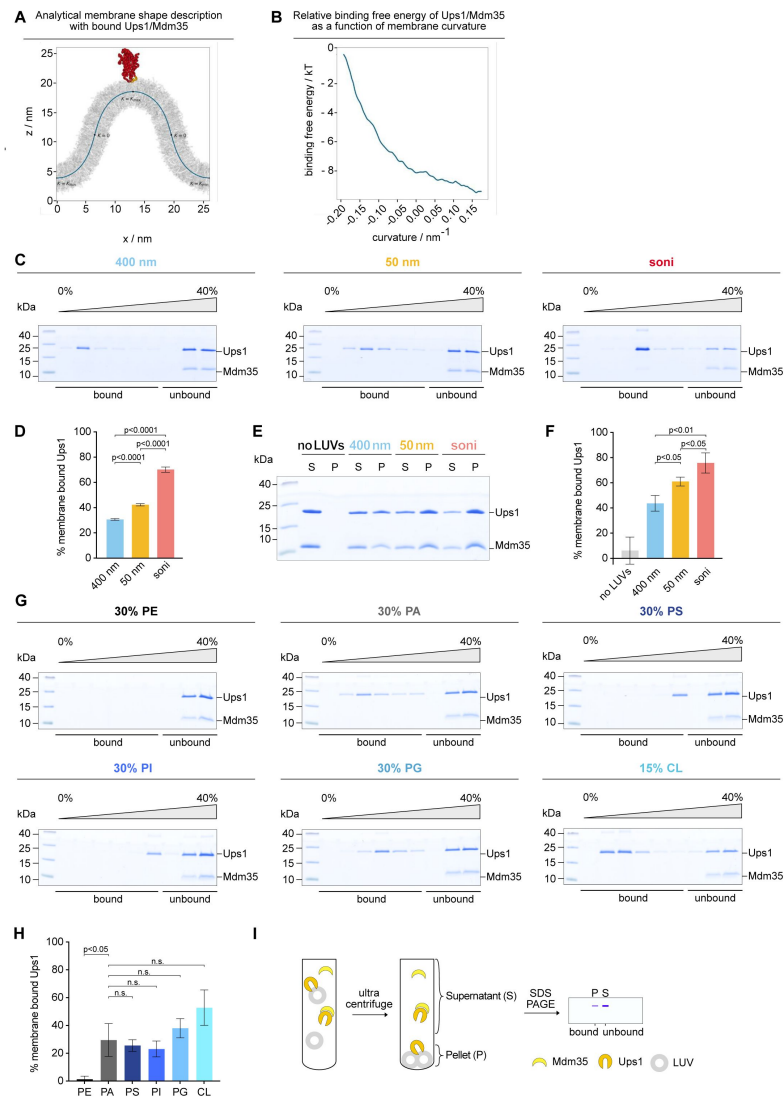


Figure 2.

Ups1 membrane binding is dependent on curvature and electrostatic interactions.

(A) Schematic illustration of the coarse-grained molecular dynamics simulation setup used to investigate the curvature-dependent binding of Ups1 to lipid membranes. The analytical membrane shape description is shown together with the Ups1/Mdm35 complex bound to the membrane. (B) Relative binding free energy ($\Delta\Delta F$) of Ups1/Mdm35 as a function of membrane curvature. (C) Flotation analyses showing the binding of untagged Ups1/Mdm35 to liposomes extruded through membranes with indicated pore sizes and composed of 50% PC, 30% PA, 19.95% PE, 0.05% TF488-PE at pH 7.0. A representative image from three independent experiments is shown for each condition. (D) The percentage of membrane-bound Ups1 relative to total Ups1 quantified from Coomassie-stained gels of (C). Data represent the mean $\pm$ SD from three independent experiments. (E) Co-sedimentation analyses showing the binding of untagged Ups1/Mdm35 to liposomes extruded through membranes with the indicated pore sizes. Liposomes were composed of 50% PC, 30% PA, 19.95% PE and 0.05% TF488-PE at pH 7.0. A representative image from three independent experiments is shown for each condition. (F) The percentage of membrane-bound Ups1 relative to total Ups1 quantified from Coomassie-stained gels of (E). Data represent the mean $\pm$ SD from three independent experiments. (G) Flotation analyses showing the binding of untagged Ups1/Mdm35 to liposomes composed of 69.875% PC, 0.125% TF488-PE and 30% of either PE, PA, PS, PI, or PG, as indicated. CL liposomes were composed of 84.875% PC, 0.125% TF488-PE, and 15% CL. Liposomes were extruded through 50 nm membranes and the assay was performed at pH 7.0. A representative image from three independent experiments is shown for each condition. (H) The percentage of membrane-bound Ups1 relative to total Ups1 quantified from Coomassie-stained gels of (G). Data represent the mean $\pm$ SD from three independent experiments. (I) Schematic diagram of the co-sedimentation assay used to study protein membrane interactions (details see Methods).

findings align well with our molecular dynamics simulations, collectively demonstrating that Ups1 exhibits a distinct binding preference for positively curved membrane regions. Interestingly, membrane deformation experiments indicate that Ups1 not only senses membrane curvature but also exhibits mild curvature-inducing activity (**Supplementary Figure 2A&B** [↗](#)). This suggests that, beyond associating with locally curved membrane regions, Ups1 might subsequently stabilize or even induce or further enhance membrane curvature.

Curvature enhances membrane association of Ups1 but does not independently drive its binding

It has been established that Ups1 binding to membranes depends on the presence of negatively charged lipids in the membrane. This conclusion is based on experiments with large liposomes, focusing on its interaction with relatively flat membranes (Connerth et al., 2012 [↗](#)). Our findings demonstrate that, in addition to charge, membrane curvature contributes substantially to Ups1 binding. However, while curvature appears to enhance membrane association, it remains unclear whether it is sufficient to drive binding independently.

To address this, we analyzed Ups1 binding to highly curved neutral membranes and compared its binding to highly curved negatively charged membranes. No Ups1 binding was observed to neutral highly curved membranes, indicating that curvature alone is insufficient to support Ups1 binding (**Figure 2G&H** [↗](#)).

Ups1 bound to highly curved negatively charged membranes, with no clear preference for a specific negatively charged lipid species (**Figure 2G&H** [↗](#)). Compared to PA, CL showed a trend toward stronger binding, however this effect was not statistically significant under our experimental conditions. Thus, our findings significantly extend previous results obtained using large vesicles at pH 5.5, which demonstrated a strict requirement for negative charges and increased binding to CL-containing membranes. This dependency is not only maintained in highly curved membranes but also persists at the physiological pH of 7.0. Taken together, these findings suggest that membrane curvature plays a crucial role in promoting Ups1 binding to membranes and that it acts in concert with electrostatic interactions rather than serving as an independent determinant of membrane association.

Curvature enhances Ups1 binding to the MOM, while the high negative charge density of the MIM supports strong binding with a lesser curvature dependency

The two mitochondrial membranes have a distinct and complex lipid composition. The MIM contains relatively high levels of CL, which contributes with its two negative charges to the high negative charge density of the MIM. In contrast, the MOM contains less CL and also less other negatively charged lipids, resulting in a membrane with less overall surface charge (Zinser et al., 1991 [↗](#)). To gain insight into whether curvature sensing is relevant for Ups1 membrane binding *in vivo*, we assessed the curvature-dependent binding of Ups1 to membranes mimicking the native lipid compositions of both, MOM and MIM. Binding to the MOM was significantly influenced by curvature, with 45% of Ups1 binding to flat membranes compared to 69% on highly curved membranes (**Figure 3A** [↗](#)). This suggests that within the lipid environment of the MOM, Ups1 preferentially associates with membrane regions exhibiting positive membrane curvature. In contrast, binding to the MIM was consistently strong, reaching 86% even on flat membranes (**Figure 3B** [↗](#)), while a trend toward increased binding to highly curved MIM membranes was maintained. The strong binding of Ups1 to even flat, highly negatively charged membranes suggests that a high density of negative charges can, at least partially, override its curvature-dependent binding, enabling strong association regardless of membrane shape.

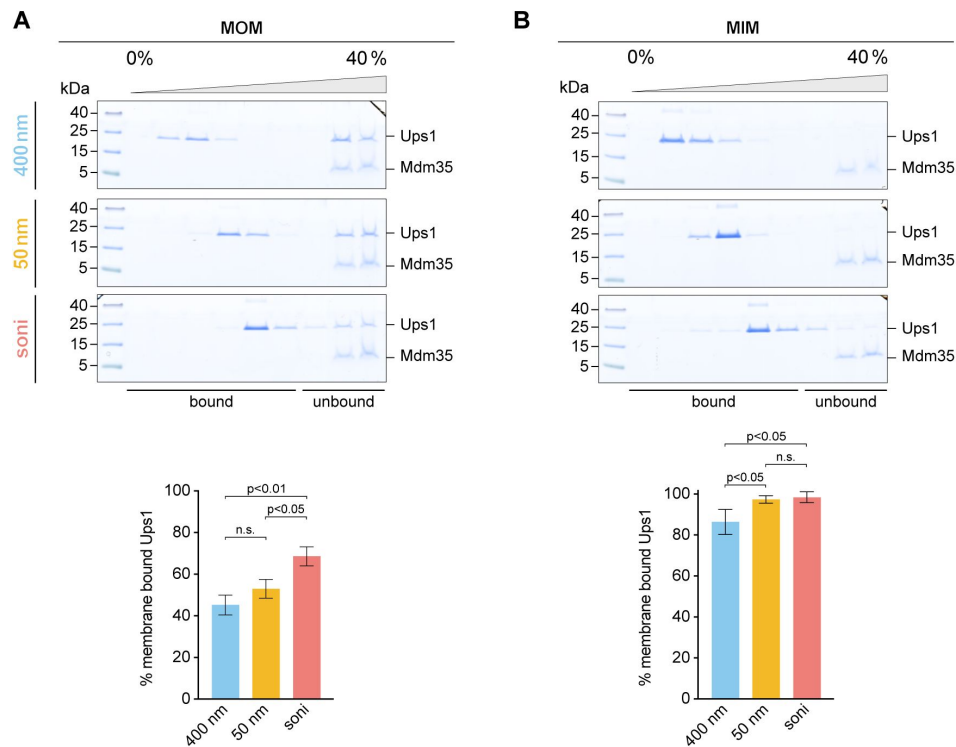


Figure 3.

Ups1 binding to membranes mimicking the MOM is curvature-dependent.

(A) Flotation analyses showing the binding of untagged Ups1/Mdm35 to liposomes extruded through membranes with indicated pore sizes, representing MOM-like membranes (46% PC, 32.95% PE, 10% PI, 1% PS, 6% CL, 4% PA, 0.05% TF488-PE) at pH 7.0. The percentage of membrane-bound Ups1 relative to total Ups1 was quantified from Coomassie-stained gels. Data represent the mean $\pm$ SD from three independent experiments. **(B)** Flotation analyses showing the binding of untagged Ups1/Mdm35 to liposomes representing MIM-like membranes (38% PC, 23.95% PE, 16% PI, 4% PS, 16% CL, 2% PA, 0.05% TF488-PE). The experiment and subsequent analysis were performed as in (A).

Positive curvature increases the free energy of PA, lowering the barrier for lipid extraction

Since we observed a strong curvature-dependent binding of Ups1 to liposomes mimicking the MOM, where lipid extraction occurs, we sought to investigate whether this curvature preference contributes to the efficiency or regulation of the extraction process. Regions of positive membrane curvature, such as the outer leaflet of vesicles, exhibit increased lipid packing defects and surface tension, which enhance the exposure of alkyl chains to solvents (Risselada & Marrink, 2009 [↗](#); Vanni et al., 2014 [↗](#)). This energetically elevated state could make lipid extraction more favorable. To test this hypothesis, we employed an *in silico* approach using coarse-grained molecular dynamics simulations. Using the same buckled membrane previously used for analyzing Ups1 binding, we now focused on lipid sorting behavior (**Figure 4A** [↗](#)). Curvature-driven lipid partitioning was analyzed, revealing distinct preferences, with PE and PA preferentially localizing to negatively curved regions, while PC accumulated in positively curved regions (**Figure 4B** [↗](#)). Due to differences in initial concentrations, absolute changes in lipid fractions appear small for PA. However, the relative redistribution of PA along the curvature gradient is substantial, with a decrease of 23% in positively curvature compared to flat membranes, emphasizing its curvature-dependent sorting behavior. This overall sorting pattern aligns with the properties of the lipid headgroups. Compared to PC, PE and PA have smaller headgroups, making localization of these lipids to positively curved regions less favorable due to a reduced ability to maintain optimal packing. For PA, electrostatic repulsion may also contribute, as its curvature preference is less pronounced compared to PE, despite having an even smaller headgroup.

To determine the free energy profile of the lipids along the buckled membrane, we constructed the potential of mean force (PMF) from the lipid distributions along the reaction coordinate,

$$\text{PMF}(s) = k_B T \ln \left[\frac{\phi(s)}{\phi(s=0.25)} \right] \quad (1.1)$$

with the Boltzmann constant k_B , the temperature T , the arc length parameter s , and the lipid distribution $\phi(s)$. For PA and PE, the PMF increased from negatively to positively curved membrane regions, indicating that PA and PE are in a higher energy state in positively curved compared to flat membranes. PC followed the opposite trend (**Figure 4C&D** [↗](#)).

If a lipid is extracted from a region where it is already in a high-energy state, the energy that must be overcome is lower compared to a lipid in low energy state. Consequently, because PA has a higher free energy in positively curved regions (**Figure 4C&D** [↗](#)), the relative energy cost for extraction is reduced in these areas (**Figure 4E** [↗](#)).

Overall, our *in silico* results suggest that PA extraction by Ups1 requires less energy from regions of high positive curvature, enhancing the efficiency of lipid extraction and possibly facilitating subsequent transport from these membrane regions.

Lipid transfer is curvature-dependent, with increased efficiency driven by faster extraction from highly curved membranes

To experimentally test whether membrane curvature has a regulatory effect on Ups1-dependent lipid transfer, we employed a lipid transfer assay (Miliara et al., 2015 [↗](#); Watanabe et al., 2015 [↗](#)). In this assay, donor liposomes contain NBD-PA, which is quenched by Rhod-PE in the same membrane. Acceptor liposomes are devoid of any fluorescently labeled lipids. When NBD-PA is transferred from donor to acceptor liposomes, it becomes dequenched, allowing lipid transfer to be monitored as an increase in NBD fluorescence over time (**Figure 5A** [↗](#)). The initial linear

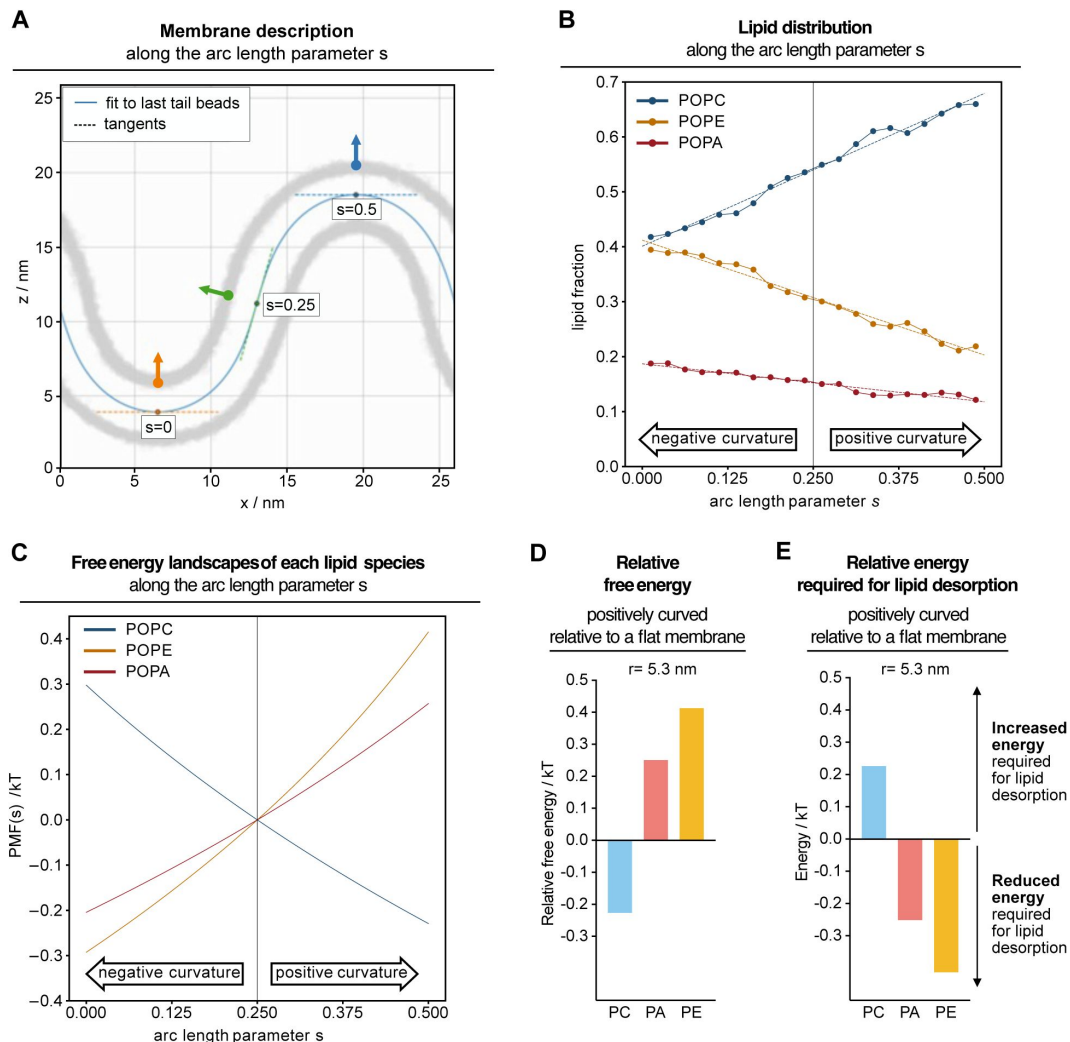


Figure 4.

Curvature-dependent lipid partitioning and free energy landscapes reveal energetically favourable PA Extraction from positively curved membranes.

(A) Schematic illustration of the coarse-grained molecular dynamics simulation setup used to analyse curvature-dependent lipid partitioning. The analytical membrane shape description is shown along the arc length parameter s . **(B)** Lipid distributions along the arc length parameter s after the system was equilibrated to allow curvature-induced lipid partitioning. Errors from bootstrapping are smaller than the symbols. Dashed lines represent the mean of linear fits to every resampled lipid distribution. **(C)** Free energy landscapes of each lipid species along the arc length parameter s constructed from the mean of linear fits to the lipid distributions. **(D)** Relative free energy difference of the indicated lipid species in a positively curved compared to a flat membrane. **(E)** Relative energy differences required for lipid desorption from a positively curved compared to a flat membrane.

increase in fluorescence reflects the transfer velocity. The gradual subsequent slowing of the fluorescence increase indicates that back-transfer becomes relevant, such that not all lipid transfer events contribute to the fluorescence increase.

To analyze the impact of donor and acceptor size on lipid transfer by Ups1, we compared sonicated, 50 nm, and 400 nm extruded liposomes. Liposomes without the addition of protein resulted in a negligible fluorescence increase, whereas the addition of Ups1 led to a clear increase, indicating successful lipid transfer (**Figure 5B&C**). Notably, transfer efficiency was strongly influenced by vesicle size (**Figure 5C**). When both donor and acceptor liposomes were small, transfer velocity increased 14-fold compared to relatively flat 400 nm vesicles.

To assess the effect of donor size, we performed experiments where the acceptor size was kept constant while the donor size varied (**Figure 5D&E**). When sonicated acceptor liposomes were used, transfer velocities significantly and gradually increased with higher donor curvature. Similar results were observed for 50 nm and 400 nm extruded acceptor vesicles. This shows that donor size, and thus the membrane curvature from which the lipid is extracted, is a strong determinant of transfer velocity. Interestingly, acceptor size had a lesser impact, as shown by experiments with a constant donor and varying acceptor size (**Figure 5F&G**). When the donor liposomes were 50 nm or 400 nm, varying the acceptor size had only a minor impact on transfer velocity. When donor liposomes were highly curved (sonicated), a moderate effect of acceptor size on lipid transfer was observed. This shows that insertion efficiency becomes relevant only when extraction occurs very rapidly, demonstrating that extraction is the rate-determining step while insertion plays a minor role unless extraction is extremely fast.

To further validate that extraction is curvature-dependent, we performed an extraction assay (Watanabe et al., 2015). Donor liposomes containing NBD-PA and Rhod-PE were incubated with Ups1 in the absence of acceptor liposomes (**Figure 5H**). Under these conditions, the fluorescence increase originates exclusively from NBD-PA uptake into Ups1. Consistent with the lipid transfer results, extraction efficiency increased with decreasing donor size (**Figure 5I**). Extraction was 5-fold faster from sonicated compared to 400 nm liposomes. Taken together, this confirms that lipid extraction is strongly influenced by membrane curvature and is more efficient from highly curved membranes.

Discussion

In this study, we show that Ups1 membrane binding is regulated by a combination of electrostatic interactions and membrane curvature to optimize lipid transfer efficiency. While electrostatic interactions were previously proposed to mediate Ups1 binding, this was based on experiments conducted at pH 5.5 with His-tagged Ups1 (Connerth et al., 2012), where the positive charge of the His-tag could have influenced binding to negatively charged membranes. We now demonstrate that, even in the absence of the His-tag and at a physiological pH of 7.0, Ups1 binding is dependent on negatively charged lipids, indicating electrostatic interactions with positively charged residues in the protein. Our data further reveal that Ups1 does not exhibit a strict preference for a specific lipid headgroup but rather recognizes membrane charge more broadly. Previous studies suggested a preference for CL, which carries two negative charges (Connerth et al., 2012). These results may have resulted from an overall increase in membrane charge when comparing the same molar amounts of CL with other negatively charged lipids. We maintained a constant total negative charge across lipid compositions by using only half the amount of CL compared to other singly negatively charged lipids. Nevertheless, we observed a trend toward increased binding to CL, though it did not reach statistical significance. Given the proposed role of CL in regulating Ups1 via negative feedback (Acoba et al., 2020; Connerth et al., 2012), the possibility of differential binding to CL compared to other lipids warrants further investigation. However, it should be noted that CL is likely enriched in specific sub-domains within the MIM and might not be readily

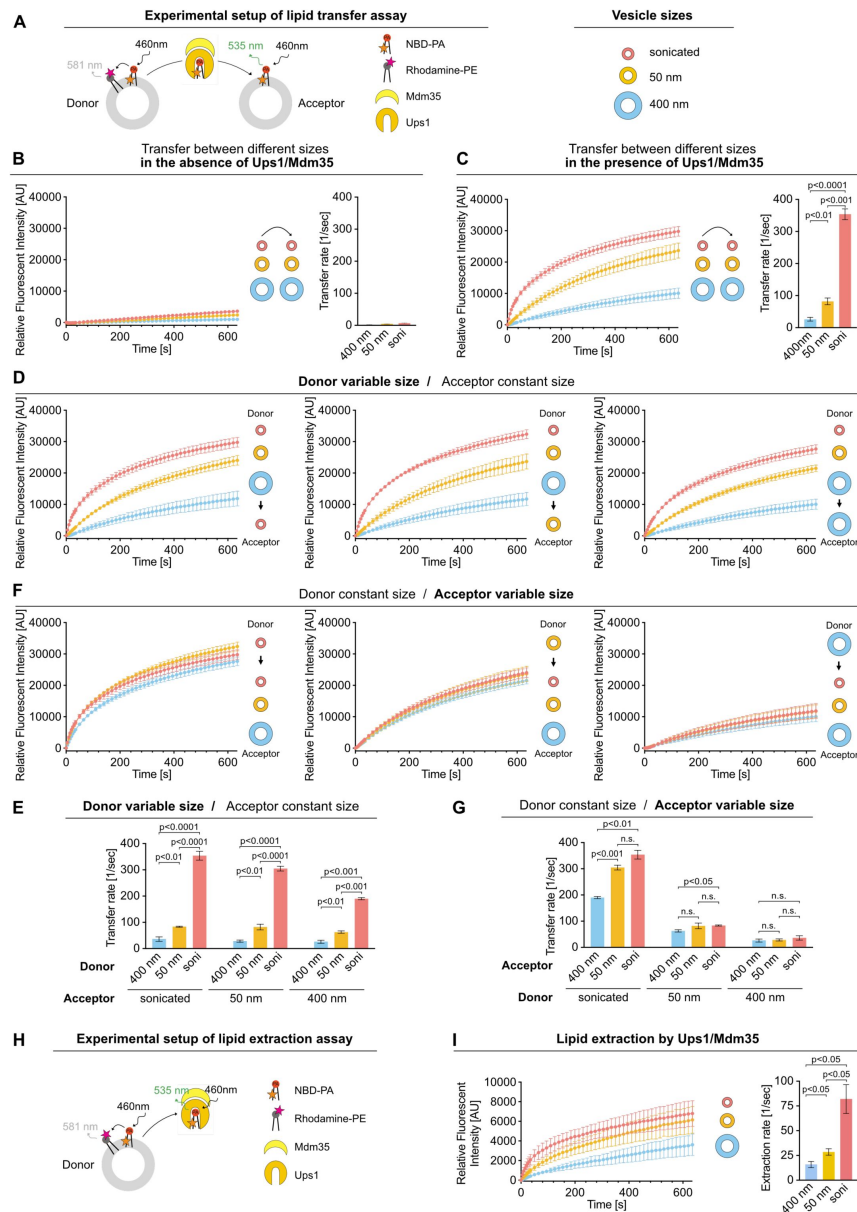


Figure 5.

Lipid Transfer by Ups1/Mdm35 Depends on Donor Curvature.

(A) Schematic diagram of the transfer assay to study Ups/Mdm35 mediated PA transfer. (B-G) PA transfer between donor and acceptor liposomes extruded through membranes with the indicated pore sizes. Donor liposomes contained 8% NBD-PA, 2% Rhod-PE, 90% DOPC and acceptor liposomes 10% DOPA and 90% DOPC. (B) Size dependent spontaneous PA transfer between donor and acceptor liposomes in the absence of Ups1/Mdm35. Normalized NBD fluorescence is shown over time. Data represent the mean $\pm$ SD of three independent replicates (left). The transfer rate is determined from the slope of the linear NBD increase using linear regression (right). Data represent the mean $\pm$ SE of three independent replicates. (C) Size dependent PA transfer between donor and acceptor liposomes in the presence of Ups1/Mdm35. Same experimental setup and analyses as in (B), with Ups1/Mdm35 added at time point 0. (D+E) Effect of donor size on PA transfer by Ups1/Mdm35. Same experimental setup and analyses as in (C). (F+G) Effect of acceptor size on PA Transfer by Ups1/Mdm35. Same experimental setup and analyses as in (C). (H) Schematic diagram of the extraction assay to study Ups1/Mdm35 mediated PA extraction from donor liposomes. (I) PA extraction from donor liposomes with Ups1/Mdm35 added at time point 0. Normalized NBD fluorescence is shown over time. Data represent the mean $\pm$ SD of three independent replicates (left). The extraction rate is determined from the slope of the linear NBD increase using linear regression (right). Data represent the mean $\pm$ SE of three independent replicates.

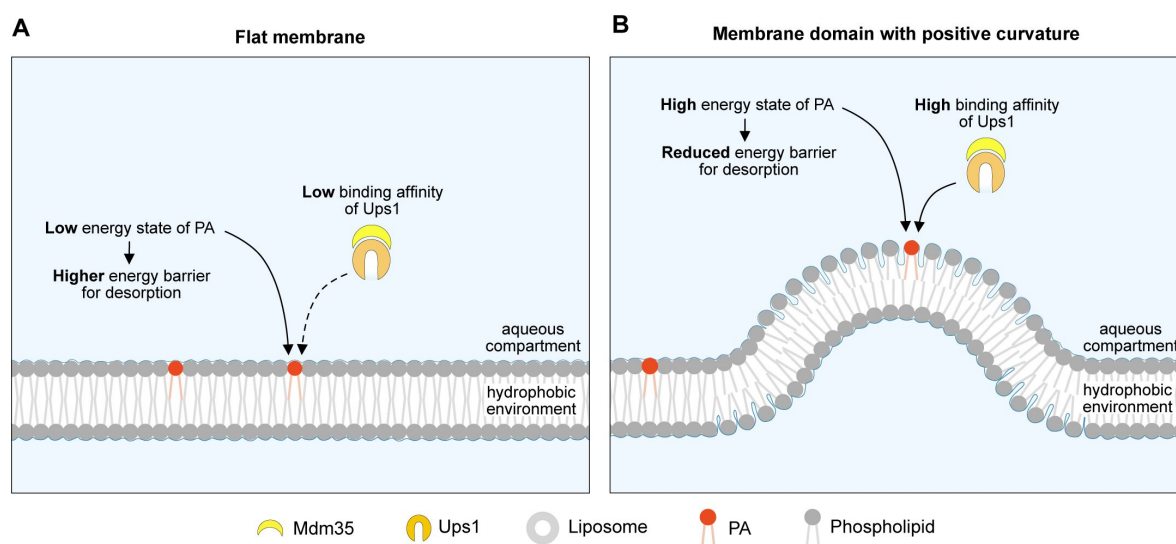


Figure 6.

Curvature-Dependent PA Binding and Its Role in Efficient PA Extraction by Ups1

(A) Ups1 has low binding affinity to flat membranes, where PA is in a low-energy state, making extraction energetically unfavourable. **(B)** In contrast, Ups1 preferentially binds to highly positively curved membranes, where PA is in a high-energy state, lowering the energy barrier for extraction and enabling more efficient lipid transfer.

available as a binding platform at the IMS side of the MIM. Due to its geometry and characteristics of a non-bilayer lipid it will likely sort into regions of negative curvature (Decker & Funai, 2024 [DOI](#); Elias-Wolff et al., 2019 [DOI](#)). Furthermore, it is known to stabilize and, most likely, closely interact with several MIM protein complexes (Duncan et al., 2018 [DOI](#); Lange et al., 2001 [DOI](#); Musatov & Sedlak, 2017 [DOI](#)), making it unlikely to be evenly distributed across the plane of the MIM and between both bilayer leaflets. Thus, the extent to which CL regulates Ups1 function remains to be determined.

Our findings furthermore demonstrate that membrane binding by Ups1 is not solely dictated by electrostatic interactions but is significantly influenced by membrane morphology. Ups1 shows a clear preference for positively curved over flat or negatively curved membranes. However, high curvature alone is insufficient to promote binding, as negatively charged lipids are still required for detectable interaction with highly curved membranes. Interestingly, the dependency on curvature can be largely overridden by highly negatively charged membranes, as observed in our analysis of mitochondrial inner membrane lipid compositions. Under these conditions, Ups1 binding was nearly complete even on flat membranes. Taken together, curvature sensing appears to fine-tune Ups1 binding affinity rather than being the primary determinant of membrane interaction.

For physiological membrane compositions and at the physiological pH of the IMS Ups1 curvature dependent binding was especially apparent for the MOM where PA extraction takes place. Notably, molecular dynamics simulations suggest that phosphatidic acid extraction is energetically more favorable from positively curved membranes. Our lipid transfer experiments are in agreement with this model, as PA extraction was significantly more efficient from highly curved membranes compared to flat membranes. This correlated with a substantial increase in lipid transfer rates. Based on our findings, we propose a model in which Ups1 preferentially binds to positively curved membrane regions where PA exists in an energetically unfavorable state, allowing for efficient extraction. This spatial preference ensures that Ups1 engages with membrane domains where extraction is energetically favorable, thereby increasing lipid transfer efficiency. This mechanism is particularly relevant in environments dominated by flat membranes, such as the MOM, where it helps optimize the use of Ups1 for effective lipid transport.

The molecular details by which Ups1 senses membrane curvature remain unclear. Various mechanisms have been described for curvature sensing, including scaffolding and hydrophobic insertion (Has & Das, 2021 [DOI](#); Johnson et al., 2024 [DOI](#)). Scaffolding relies on a binding interface that conforms to membrane curvature, thereby maximizing the contact area between favorable protein-membrane contact sites. This mechanism is highly specific to the geometry of the membrane and the protein. From the known structures of Ups1/PRELID1, though all solved in solution and not in contact with a membrane, it seems unlikely that this mechanism is used. In contrast, hydrophobic insertion is driven by lipid-packing defects induced, for example, by curvature, where amphipathic helices or hydrophobic protrusions insert into the bilayer to stabilize membrane interaction. This mode of binding is not necessarily specific to curvature itself but rather to the packing defects that arise from curvature. Determining which mechanism underlies Ups1 curvature sensing will provide insight into whether Ups1 specifically detects curvature or whether it more generally responds to changes in membrane packing. This distinction is crucial, as it would indicate an additional level of regulation in which Ups1 binding and lipid transfer are influenced not only by membrane charge and curvature but also by other determinants of the membrane that induce packing defects such as acyl chain composition (Vanni et al., 2014 [DOI](#)).

To gain deeper insight into how Ups1 binds to membranes and senses membrane curvature, it is essential to identify the structural elements responsible for these functions. Curvature sensing is likely an intrinsic property of its membrane-binding interface; however, the lack of structural data for membrane-bound Ups1 makes it challenging to determine the precise molecular features

involved. A conformational change upon membrane binding seems likely, as indicated by the fact that Mdm35 must dissociate for Ups1 to efficiently associate with membranes, even though its binding site is distinct from the membrane-binding interfaces predicted by molecular dynamics simulations (Lu et al., 2020; Watanabe et al., 2015). Understanding the conformation of membrane-associated Ups1 will be essential for elucidating the molecular basis of its membrane interactions, including the role of negative charges, potential headgroup specificity—such as a preference for CL—and the mechanisms governing its curvature-dependent association.

Understanding the conformation of membrane-associated Ups1 might also contribute to elucidating the molecular basis of its observed pH-dependent binding to membranes. Since the most significant changes in Ups1 binding with respect to pH align with the physiological pH range of the mitochondrial intermembrane space (IMS), it is an interesting question whether this pH dependency plays a regulatory role in Ups1 function in vivo. Notably, the pH of the IMS fluctuates depending on cellular conditions such as mitochondrial respiration and evidence suggests the presence of a lateral pH gradient (Rieger et al., 2014; Toth et al., 2020). While the presence of a pH gradient across the IMS remains debated, it is plausible that the MIM surface, where protons are pumped by the OXPHOS complexes, could experience a lower pH than the MOM surface, which is thought to be more equilibrated with the cytosol due to its porins (Benz, 1990). The pH range at which Ups1 undergoes its binding transition suggests that temporal or localized fluctuations in IMS pH could directly modulate Ups1 membrane binding. This dynamic regulation may contribute to establishing directionality or to fine-tune Ups1 activity in response to physiological conditions.

Taken together, our findings demonstrate that the activity of Ups1/Mdm35 is highly sensitive to the lipid composition and morphology of mitochondrial membranes, as well as to pH. Since CL, a central lipid for mitochondrial function, depends on efficient PA transfer by Ups1/Mdm35 for its synthesis, these observations provide insights into how and why alterations in membrane morphology and lipid content may affect CL synthesis and, ultimately, mitochondrial function. Given that Ups1/Mdm35 are structurally and functionally conserved in mammals, these insights may also offer valuable perspectives for understanding the pathophysiology of mitochondrial-related diseases.

Methods

Recombinant protein expression and purification

Expression and purification of His-Ups1/Mdm35 was performed as described before with following modifications (Connerth et al., 2012). The coding sequence for *UPS1* carrying an N-terminal His-tag, and *MDM35* from *Saccharomyces cerevisiae* were cloned into the pET-Duet-1 (Merck) vector and co-expressed in Origami B(DE3)pLysS *E.coli* cells (Merck) in the presence of 1 mM IPTG at 30°C for 3h. Cells were harvested and resuspended in lysis buffer (250 mM NaCl, complete protease inhibitor cocktail without EDTA (Roche), 1 mM PMSF, 50 mM Tris/HCl pH 8.0). The lysate was centrifuged at 14,000 rpm for 30 min at 4°C and the supernatant was filtered through a 0.45 µm filter. The clarified lysate was loaded onto a HisTrap FF column (Cytiva). After a wash step with wash buffer (250 mM NaCl, 20 mM imidazole, 50 mM Tris/HCl pH 8.0), proteins were eluted with elution buffer (250 mM NaCl, 500 mM imidazole, 50 mM Tris/HCl pH 8.0) and the protein containing fractions were collected. The protein was further purified by size exclusion chromatography using a HiLoad 16/600 Superdex 75 pg (Cytiva) column with SEC buffer (150mM NaCl, 25mM HEPES/HCl pH 7.4). Fractions were analyzed by SDS-PAGE followed by Coomassie blue staining. Selected fractions were pooled and concentrated using an Amicon filter (Merck Millipore).

To obtain Ups1/Mdm35 without a His-tag, a TEV cleavage site was inserted between the His-tag and Ups1 in the previously used pET Duet vector. Expression and purification of His-TEV-Ups1/Mdm35 was performed as described above. The His-tag was removed by incubation with His-tagged TEV protease (van den Berg et al., 2006) in the presence of 3 mM glutathione and 0.3 mM oxidized glutathione. The uncleaved protein and the TEV protease were removed using a HisTrap FF column. Proteins were eluted using an imidazole gradient from 20 mM to 500 mM in 250 mM NaCl, 50 mM Tris/HCl, pH 8.0. Fractions containing the cleaved protein were pooled, rebuffed to 150 mM NaCl, 25 mM HEPES/HCl pH 7.4 and concentrated using an Amicon tube.

Liposome preparation

Liposome preparation was performed as previously described (Denkert et al., 2017). Unless stated otherwise, natural lipids were used as follows: PC (egg L- α -Phosphatidylcholine), PE (egg L- α -Phosphatidylethanolamine), PA (egg L- α -Phosphatidic acid), PS (brain L- α -Phosphatidylserine), PI (liver L- α -Phosphatidylserine) and CL (heart cardiolipin). As indicated, the following lipids were used: DOPC (18:1 (Δ 9-Cis) PC), DOPA (18:1 (Δ 9-Cis) PA), Rhodamine-PE (18:1 Liss Rhod PE), NBD-PA (18:1-12:0 NBD PA), and TF488-PE (18:1 PE-TopFluor™ AF488).

All lipids were purchased from Avanti Polar Lipids. For liposome formation, lipids were mixed in the desired molar ratios, dried briefly under a stream of nitrogen gas, and dried in a desiccator for at least 2 hours. Dried lipids were hydrated in the respective assay buffer. For the pH-dependent binding assay, assay buffers with the indicated pH were used (150 mM NaCl, 10 mM MES/NaOH, pH 5.5); (150 mM NaCl, 10 mM MES/NaOH, pH 6.0); (150 mM NaCl, 10 mM MES/NaOH, pH 6.5); (150 mM NaCl, 50 mM Tris/HCl, pH 7.0); (150 mM NaCl, 50 mM Tris/HCl, pH 7.5). Unless otherwise indicated, all other experiments were conducted in assay buffer containing 150 mM NaCl and 25 mM HEPES/HCl at pH 7.0. LUVs and SUVs were prepared through freeze-thaw cycles, followed by extrusion through polycarbonate membranes (Whatman) with pore sizes of 50 or 400 nm, respectively. To obtain sonicated liposomes, 50 nm extruded liposomes were subjected to sonication (1 min at 50% amplitude and 50% output). For flotation and co-sedimentation assays liposomes were fluorescently labeled with TF488-PE as indicated. The fluorescent intensity of liposomes before and after extrusion was measured and used to determine the liposome concentration after extrusion. For RfS measurements, the lipids PC/PE/CL/PA were mixed in the following molar ratio 55:20:10:15. After drying, SUVs were generated by swelling the lipid films in buffer (150 mM NaCl, 10 mM MES/NaOH, pH 5.5) followed by sonication.

Liposome flotation

10 μ M of the Ups1/Mdm35 complex was incubated with liposomes (5 mM total lipids) in the corresponding assay buffer (see above) for 30 min at 25–30°C in a total reaction volume of 100 μ L. The sample was then mixed with Histodenz in assay buffer to a final concentration of 40% Histodenz in a total volume of 800 μ L and transferred to the bottom of an ultracentrifuge tube. A discontinuous Histodenz gradient (40%/20%/10%/5%/0%) was created by sequentially overlaying 900 μ L each of 20%, 10%, and 5% Histodenz in assay buffer followed by assay buffer on top. The sample was subjected to ultracentrifugation at 45000 rpm (SW60 Ti rotor, Beckman Coulter) for 1 h at the same temperature as above. Nine fractions of 500 μ L each were collected, precipitated with 10% TCA, and subjected to SDS-PAGE analysis.

Co-sedimentation assay

Liposomes were mixed with 10 μ M Ups1/Mdm35 in a total reaction volume of 120 μ L in the corresponding assay buffer (see above). The sample was incubated for 30 min at room temperature. After incubation the sample was subjected to ultracentrifugation for 30 min at 55000 rpm (TLA-55 rotor, Beckman Coulter) at 25°C. The supernatant was separated from the pellet and both fractions were prepared for SDS-PAGE analysis.

Tubulation assay

1-2 μl of a lipid solution containing 2 μM lipids of the indicated composition in chloroform was dried on a glass slide. Subsequently, the lipids were rehydrated in 100 μl buffer (50 mM Tris pH 7.0, 150 mM NaCl). Formed lipid membranes were visualized by Differential interference contrast (DIC) microscopy using a CellR imaging station (Olympus) with a x 80 Objective and an ORCA-ER camera (Hamamatsu Photonics) using the Xcellence rt software (Olympus). Images were taken at the indicated timepoints before or after the addition of 10 μM of the indicated proteins and processed using Fiji.

Reflectometric interference spectroscopy (RIfS) measurements

Reflectometric interference spectroscopy experiments were performed with a home-built setup, consisting of a tungsten halogen light source (LS-1, Ocean Optics, Dunedin, Florida, USA), a flow chamber transparent for visible light which firmly covers a silicon wafer with 5 μm silicon oxide layer (Active Business Company, Brunnthal, Germany) and a spectrometer (Nanocalc-2000-UV/VIS, Ocean Optics). The detailed setup and data analysis to obtain values of the optical thickness is described elsewhere (Stephan et al., 2014 [DOI](#)).

The flow chamber was first purged with buffer containing 10mM MES, 150 mM NaCl, at pH 5.5. To create a supported membrane, the PC/PE/CL/PA (55:20:10:15) containing SUV dispersion was flushed through the system until reaching a stable plateau of the optical thickness. After rinsing of the supported membrane with buffer, the system was put into a solution of Ups1/Mdm35. In order to obtain a Langmuir isotherm, the experiment was conducted with increasing concentration of Ups1/Mdm35. Lastly, the membrane was rinsed with buffer. To analyse the RIfS data, the obtained optical thicknesses were plotted against the concentration and a Langmuir isotherm was fitted to the data according to eq. 1 [DOI](#).

$$OT = OT_{\max} \cdot K_{\text{ass}} \cdot \frac{cK_{\text{ass}}}{1+cK_{\text{ass}}} \quad (1)$$

With OT being the optical thickness, $OT_{\max}$ the equilibrium optical thickness at the maximum concentration of protein, c the protein concentration and K_{ass} the thermodynamic association constant. The thermodynamic dissociation constant (K_d) is the inverse of the association constant.

Electron Microscopy of LUVs

LUVs in the presence or absence of protein were bound to a glow discharged carbon foil covered 400 mesh grid and stained with 1% uranyl acetate solution as described before (Vasic et al., 2020 [DOI](#)). Electron microscopic imaging and visualization of the sample was performed at room temperature with a Talos L120C transmission electron microscope (ThermoFischer Scientific, Eindhoven, and The Netherlands).

In silico curvature sensing

Coarse-grained molecular dynamics simulations were performed with GROMACS 2020.3 (Abraham et al., 2015 [DOI](#)) using the Martini 2.2 force field (de Jong et al., 2013 [DOI](#); Monticelli et al., 2008 [DOI](#)). The atomistic protein structure (PDB 4XHR, (Yu et al., 2015 [DOI](#))) was translated to a coarse-grained description using the martinize Python script (de Jong et al., 2013 [DOI](#)). Protein secondary structure was determined employing DSSP (Kabsch & Sander, 1983 [DOI](#); Touw et al., 2015 [DOI](#)) and overall protein structure was stabilized via elastic band with a force constant of 500 kJ/(mol*nm²).

Initial configurations of the protein membrane system were generated with the python script insane (Wassenaar et al., 2015 [DOI](#)). 674 lipids per leaflet (371 POPC, 202 POPE, 101 POPA) were put in the x-y plane of a 40 nm * 10 nm * 20 nm simulation box. The protein complex was placed approximately 1 nm above the upper leaflet. The system was solvated with standard Martini water

and a 0.15 M NaCl concentration. Steepest-descent energy minimization and initial equilibration (10 ns NVT and 50 ns NPT) were followed by compressing the system in x-direction by applying a pressure of 3 bar in x-direction while allowing the system to expand in z-direction only (Berendsen barostat ($\tau_p=12.0$ ps, compressibility of 3×10^{-4} bar $^{-1}$) (Berendsen et al., 1984 [DOI](#))). From the compression trajectory a frame close to the chosen compression level was selected. The selected frame was then used as a starting configuration of a long equilibration phase that was 5 microseconds long (Parrinello-Rahman barostat ($\tau_p=12.0$ ps, compressibility of 3×10^{-4} bar $^{-1}$) (Parrinello & Rahman, 1981 [DOI](#))). This long run is necessary to allow curvature-induced lipid partitioning to finalize. For the umbrella sampling the buckled membrane was fixed to its analytical shape by position restraints on the headgroups of lower leaflet lipids with a small force constant (10 kJ mol $^{-1}$ nm $^{-2}$). Influence on upper leaflet dynamics is minimal with these restraints. To generate starting configurations for the umbrella sampling, the protein complex was pulled along the membrane in x-direction. 60 windows equidistant in arc length parameter s , between $s = 0$ and $s = 0.5$ were used. Umbrella sampling runs were 1.05 μ s long, with the first 0.05 μ s for equilibration. Harmonic potentials with a force constant $k = 100$ kJ mol $^{-1}$ nm $^{-2}$ were used to restrain the hydrophobic loop to a defined position along the reaction coordinate. Umbrella sampling simulations were performed in the NVT ensemble. All simulations were coupled to a constant heat bath using the velocity rescale algorithm (Bussi et al., 2007 [DOI](#)).

Lipid concentrations along the arc length parameter s were extracted from a 2 μ s simulation in the NVT ensemble. System setup was as described above, except the protein complex was not present in this simulation. Since the membrane shape is mirror symmetric around $s = 0.5$, lipid distributions are averaged over both sides and presented from $s = 0$ to $s = 0.5$. Potentials of mean force were constructed from linear fits to the lipid distributions. Errors were estimated via bootstrapping (Newman & Barkema, 1999 [DOI](#)). Each lipid distribution was resampled 1000 times.

Lipid transfer assay

PA transfer was analyzed using a fluorescence dequenching assay as described (Miliara et al., 2015 [DOI](#); Watanabe et al., 2015 [DOI](#)) with the following modifications. 3.125 μ M donor liposomes (90% DOPC, 2% Rhodamine-PE, 8% NBD-PA) and 12.5 μ M acceptor liposomes (90% DOPC, 10% DOPA) were incubated with 120 nM Ups1 at 25 °C. NBD fluorescence was recorded over time using a Pherastar BMG Labtech plate reader. The fluorescence intensity measured 0.6 s after injection was set to 0 for normalization (relative fluorescent intensity). The transfer rate was determined by linear regression analysis of the initial linear NBD increase.

pKa predictions

pKa values were predicted using PROPKA version 2.0 via the VEGA online server (Li et al., 2005 [DOI](#); Pedretti et al., 2021 [DOI](#)) based on PDB structures 4XHR, 4YTW, and 4YTX (Watanabe et al., 2015 [DOI](#); Yu et al., 2015 [DOI](#)).

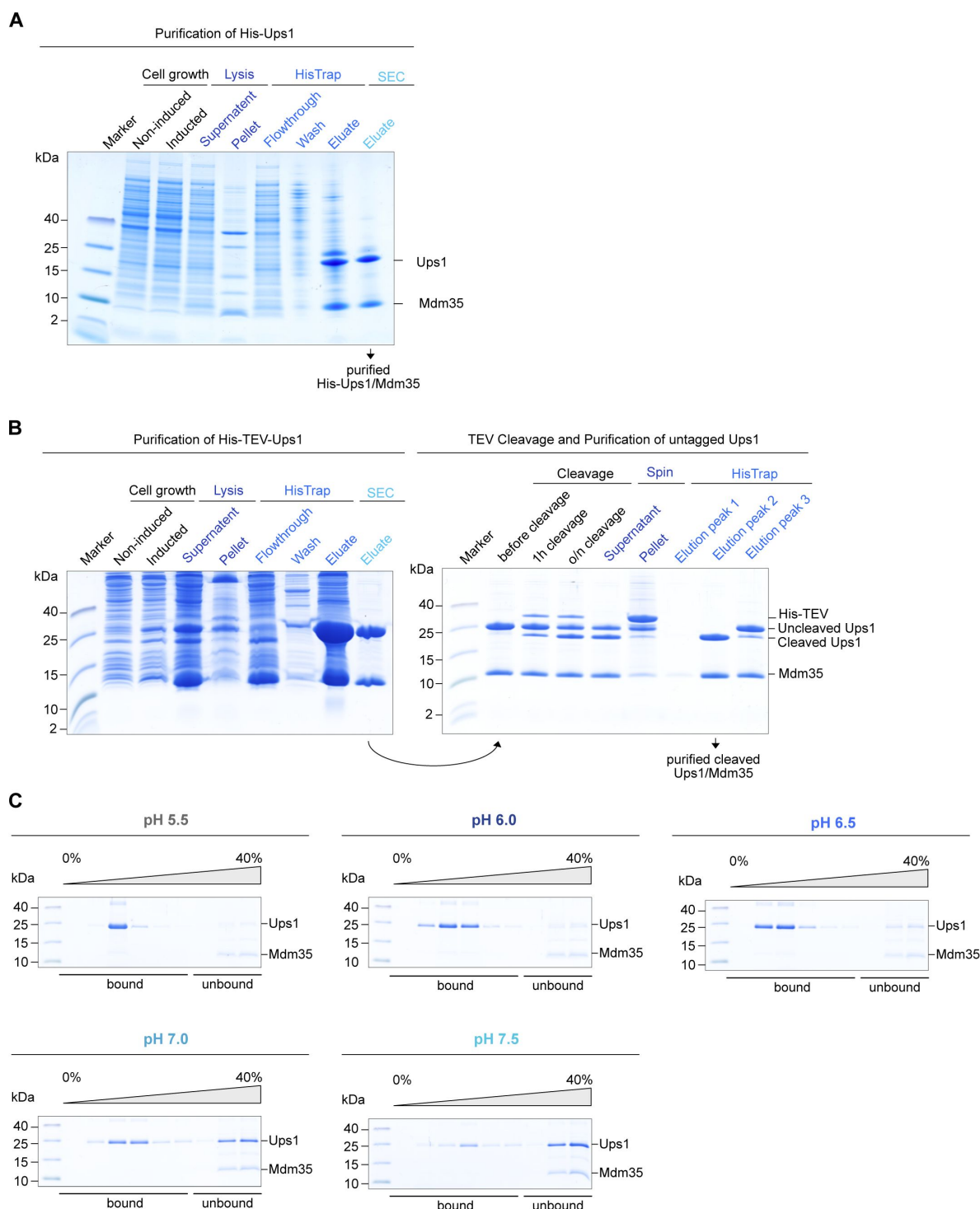
Statistical analyses

Significance was assessed using a two-tailed t-test. If the F-test to compare variances detected a significant difference, Welch's correction was applied. A significance threshold of $p < 0.05$ was used.

Use of AI tools

AlphaFold 3 (Abramson et al., 2024 [DOI](#)) was used to generate protein structural models. ChatGPT (OpenAI) was utilized for formulation refinement in the writing of this manuscript.

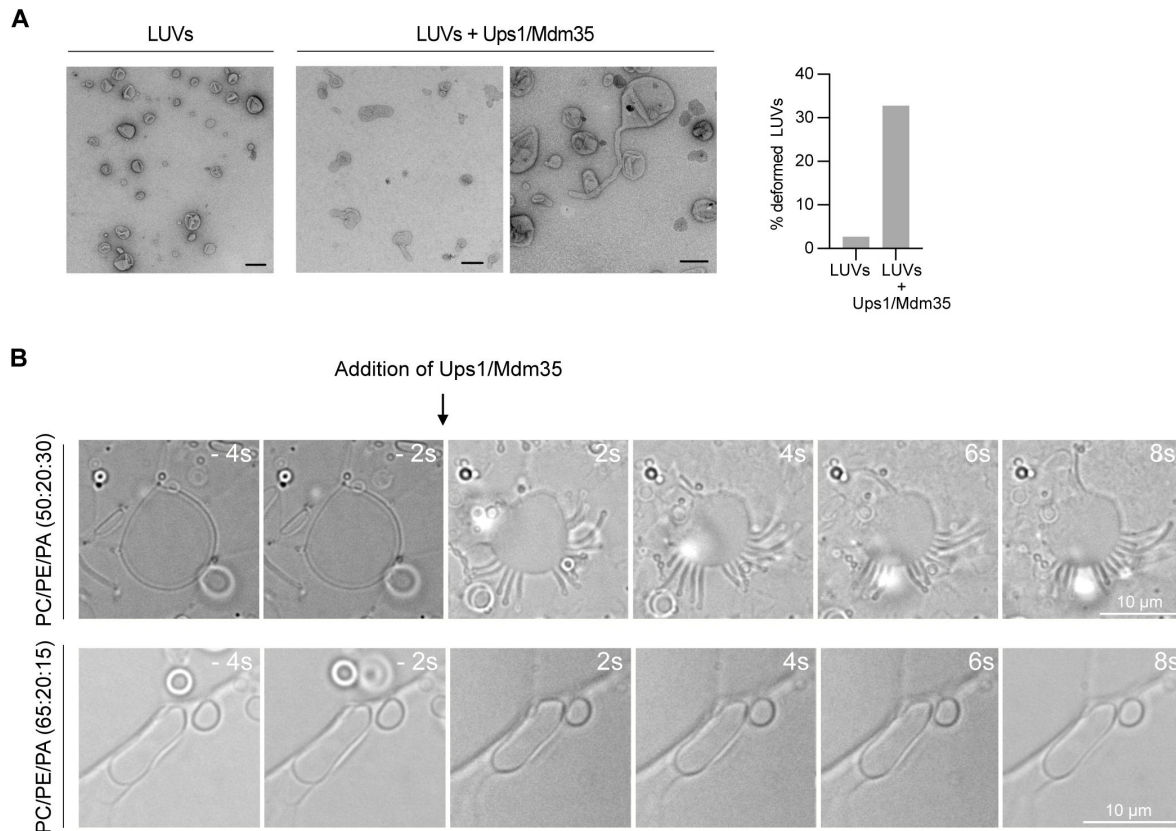
Supplementary Figures



Supplementary Figure 1.

Purification of Ups/Mdm35 and pH dependent binding of His-Ups1/Mdm35.

(A) Expression and purification of recombinant His-Ups1/Mdm35 complex analyzed by SDS-PAGE and colloidal Coomassie staining. **(B)** Expression and purification of recombinant His-TEV-Ups1/Mdm35, followed by TEV cleavage and subsequent purification of cleaved Ups1/Mdm35, analysed by SDS-PAGE and colloidal Coomassie staining. **(C)** Flotation analyses using His-Ups1/Mdm35 and LUVs composed of 50% PC, 30% PA, 19.875% PE, 0.125% TF488-PE at indicated pH values. A representative image from three independent experiments is shown for each condition.



Supplementary Figure 2.

Ups1 exhibits mild curvature-inducing activity.

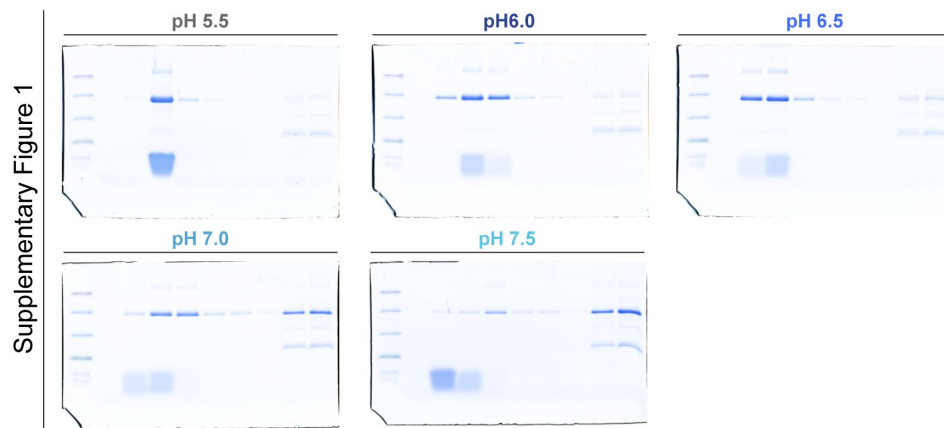
(A) Electron micrographs of liposomes (LUVs) in the absence and presence of Ups1/Mdm35 and quantification of spherical and deformed shapes in control LUVs and in LUVs after incubation with Ups1/Mdm35. A minimum of 200 membranous structures were counted per condition. Scale bar corresponds to 100 nm. **(B)** Tubulation assay in the absence and presence of Ups1/Mdm35. Indicated lipid compositions were dried on a glass slide and rehydrated using assay buffer (50 mM Tris pH 7.0; 150 mM NaCl). Formed lipid membranes were visualized by DIC imaging at the indicated timepoints before or after the addition of 10 μ M Ups1/Mdm35.



Supplementary Figure 3.

Uncropped SDS-PAGE gels.

Images of uncropped SDS-PAGE gels corresponding to the cropped gel images shown in **Figures 1** and **Supplementary Figure 1**.



Supplementary Figure 3. (continued)

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Reviewer #1 (Public review):

Lipid transfer proteins (LTPs) play a crucial role in the intramembrane lipid exchange within cells. However, the molecular mechanisms that govern this activity remain largely unclear. Specifically, the way in which LTPs surmount the energy barrier to extract a single lipid molecule from a lipid bilayer is not yet fully understood. This manuscript investigates the influence of membrane properties on the binding of Ups1 to the membrane and the transfer of phosphatidic acid (PA) by the LTP. The findings reveal that Ups1 shows a preference for binding to membranes with positive curvature. Moreover, coarse-grained molecular dynamics simulations indicate that positive curvature decreases the energy barrier associated with PA extraction from the membrane. Additionally, lipid transfer assays conducted with purified proteins and liposomes in vitro demonstrate that the size of the donor membrane significantly impacts lipid transfer efficiency by Ups1-Mdm35 complexes, with smaller liposomes (characterized by high positive curvature) promoting rapid lipid transfer.

This study offers significant new insights into the reaction cycle of phosphatidic acid (PA) transfer by Ups1 in mitochondria. Notably, the authors present compelling evidence that, alongside negatively charged phospholipids, positive membrane curvature enhances lipid transfer - an effect that is particularly relevant at the mitochondrial outer membrane. The experiments are technically robust, and my primary feedback pertains to the interpretation of specific results.

(1) The authors conclude from the lipid transfer assays (Figure 5) that lipid extraction is the rate-limiting step in the transfer cycle. While this conclusion seems plausible, it should be noted that the authors employed high concentrations of Ups1-Mdm35 along with less negatively charged phospholipids in these reactions. This combination may lead to binding becoming the rate-limiting factor. The authors should take this point into consideration. In this type of assay, it is challenging to clearly distinguish between binding, lipid extraction, and membrane dissociation as separate processes.

(2) The authors should discuss that variations in the size of liposomes will also affect the distance between them at a constant concentration, which may affect the rate of lipid transfer. Therefore, the authors should determine the average size and size distribution of liposomes after sonication (by DLS or nanoparticle analyzer, etc.).

(3) The authors use NBD-PA in the lipid transfer assays. Does the size of the donor liposomes affect the transfer of NBD-PA and DOPA similarly? Since NBD-labeled lipids are somewhat unstable within lipid bilayers (as shown by spontaneous desorption in Figure 5B), monitoring the transfer of unlabeled PA in at least one setting would strengthen the conclusion of the swap experiments.

(4) The present study suggests that membrane domains with positive curvature at the outer membrane may serve as starting points for lipid transport by Ups1-Mdm35. Is anything known about the mechanisms that form such structures? This should be discussed in the text.

<https://doi.org/10.7554/eLife.106979.1.sa3>

Reviewer #2 (Public review):

Summary:

Lipid transfer between membranes is essential for lipid biosynthesis across different organelle membranes. Ups1-Mdm35 is one of the best-characterized lipid transfer proteins, responsible for transferring phosphatidic acid (PA) between the mitochondrial outer membrane (OM) and inner membrane (IM), a process critical for cardiolipin (CL) synthesis in the IM. Upon dissociation from Mdm35, Ups1 binds to the intermembrane space (IMS) surface of the OM, extracts a PA molecule, re-associates with Mdm35, and moves through the aqueous IMS to deliver PA to the IM. Here, the authors analyzed the early steps of this PA transfer - membrane binding and PA extraction - using a combination of in vitro biochemical assays with lipid liposomes and purified Ups1-Mdm35 to measure liposome binding, lipid transfer between liposomes, and lipid extraction from liposomes. The authors found that membrane curvature, a previously overlooked property of the membrane, significantly affects PA extraction but not PA insertion into liposomes. These findings were further supported by MD simulations.

Strengths:

The experiments are well-designed, and the data are logically interpreted. The present study provides an important basis for understanding the mechanism of lipid transfer between membranes.

Weaknesses:

The physiological relevance of membrane curvature in lipid extraction and transfer still remains open.

<https://doi.org/10.7554/eLife.106979.1.sa2>

Reviewer #3 (Public review):

The manuscript by Sadeqi et al. studies the interactions between the mitochondrial protein Ups1 and reconstituted membranes. The authors apply synthetic liposomal vesicles to investigate the role of pH, curvature, and charge on the binding of Ups1 to membranes and its ability to extract PA from them. The manuscript is well written and structured. With minor exceptions, the authors provide all relevant information (see minor points below) and reference the appropriate literature in their introduction. The underlying question of how the energy barrier for lipid extraction from membranes is overcome by Ups1 is interesting, and the data presented by the authors could offer a valuable new perspective on this process. It is also certainly a challenging in vitro reconstitution experiment, as the authors aim to

disentangle individual membrane properties (e.g., curvature, charge, and packing density) to study protein adsorption and lipid transfer. I have one major suggestion and a few minor ones that the authors might want to consider to improve their manuscript and data interpretation:

Major Comments:

The experiments are performed with reconstituted vesicles, which are incubated with recombinant protein variants and quantitatively assessed in flotation and pelleting assays. According to the Materials and Methods section, the lipid concentration in these assays is kept constant at 5 μ M. However, the authors change the size of the vesicles to tune their curvature. Using the same lipid concentration but varying vesicle sizes results in different total vesicle concentrations. Moreover, larger vesicles (produced by freeze-thawing and extrusion) tend to form a higher proportion of multilamellar vesicles, thus also altering the total membrane area available for binding. Could these differences in the experimental system account for the variation in binding? To address this, the authors would need to perform the experiments either under saturation (excess protein) conditions or find an experimental approach to normalize for these differences.

<https://doi.org/10.7554/eLife.106979.1.sa1>

Author response:

Reviewer #1:

Lipid transfer proteins (LTPs) play a crucial role in the intramembrane lipid exchange within cells. However, the molecular mechanisms that govern this activity remain largely unclear. Specifically, the way in which LTPs surmount the energy barrier to extract a single lipid molecule from a lipid bilayer is not yet fully understood. This manuscript investigates the influence of membrane properties on the binding of Ups1 to the membrane and the transfer of phosphatidic acid (PA) by the LTP. The findings reveal that Ups1 shows a preference for binding to membranes with positive curvature. Moreover, coarse-grained molecular dynamics simulations indicate that positive curvature decreases the energy barrier associated with PA extraction from the membrane. Additionally, lipid transfer assays conducted with purified proteins and liposomes in vitro demonstrate that the size of the donor membrane significantly impacts lipid transfer efficiency by Ups1-Mdm35 complexes, with smaller liposomes (characterized by high positive curvature) promoting rapid lipid transfer.

This study offers significant new insights into the reaction cycle of phosphatidic acid (PA) transfer by Ups1 in mitochondria. Notably, the authors present compelling evidence that, alongside negatively charged phospholipids, positive membrane curvature enhances lipid transfer - an effect that is particularly relevant at the mitochondrial outer membrane. The experiments are technically robust, and my primary feedback pertains to the interpretation of specific results.

(1) The authors conclude from the lipid transfer assays (Figure 5) that lipid extraction is the rate-limiting step in the transfer cycle. While this conclusion seems plausible, it should be noted that the authors employed high concentrations of Ups1-Mdm35 along with less negatively charged phospholipids in these reactions. This combination may lead to binding becoming the rate-limiting factor. The authors should take this point into consideration. In this type of assay, it is challenging to clearly distinguish between binding, lipid extraction, and membrane dissociation as separate processes.

We thank the reviewer for the constructive and positive evaluation of our manuscript. We agree that, while our data support the interpretation that the rate-limiting step occurs at the donor membrane, it is difficult to dissect in our assay which of the individual steps at the donor membrane - such as binding of Ups1, lipid extraction into the binding pocket, or dissociation of Ups1 - is rate-limiting. Nevertheless, although we cannot exclude contributions from membrane binding or dissociation, several observations suggest that lipid extraction is a rate-limiting step under our experimental conditions.

The acceptor membrane has a similar lipid composition to the donor membrane (in tendency, the donor membrane is even a bit richer in binding-promoting lipids). If binding was rate-limiting, similar constraints would be expected at the acceptor membrane during lipid insertion. However, this is not observed.

Regarding dissociation, if this step were rate-limiting, one would expect similar constraints to be evident at the acceptor vesicles as well. Nevertheless, membrane dissociation might be mechanistically coupled to lipid extraction and thus difficult to evaluate as an independent step.

Based on our data and the considerations described above, we suggest that lipid extraction is the dominant rate-limiting step at the donor membrane under our conditions. However, we agree that a clear separation of these individual steps is not possible with the current experimental design. We will revise the corresponding passage to clarify that the rate-limiting step occurs at the donor membrane and, based on our observations, likely involves lipid extraction. Future studies aiming on dissecting these steps, will be important for elucidating the mechanism and regulation of Ups1-mediated lipid transfer both in vitro and in vivo.

(2) The authors should discuss that variations in the size of liposomes will also affect the distance between them at a constant concentration, which may affect the rate of lipid transfer. Therefore, the authors should determine the average size and size distribution of liposomes after sonication (by DLS or nanoparticle analyzer, etc.)

We agree that variations in liposome size will influence the average distance between vesicles at a given lipid concentration, which may in turn affect the rate of lipid transfer. As suggested, we will include DLS measurements to characterize the size distribution of our different liposome preparations.

Our setup was designed to keep the total membrane surface area comparable across conditions. This approach ensures a comparable overall binding capacity for Ups1 and enables the comparison of membrane binding and lipid extraction from different membranes. However, we agree that vesicle spacing, which is affected by liposome size at constant lipid concentration, could potentially influence certain steps in the transfer process, such as the time required for Ups1 to travel between donor and acceptor membranes. Whether this intermembrane travel time contributes to rate limitation is indeed an interesting question, and we will address this point through further discussion in the revised manuscript.

Investigating such effects in our current experimental system would require altering the vesicle concentration, which would in turn change the total membrane surface area and introduce additional variables. Nevertheless, exploring the influence of vesicle spacing and intermembrane distance on lipid transfer represents a promising direction for future studies aimed at dissecting the rate-limiting steps of the transfer cycle.

(3) The authors use NBD-PA in the lipid transfer assays. Does the size of the donor liposomes affect the transfer of NBD-PA and DOPA similarly? Since NBD-labeled lipids are

somewhat unstable within lipid bilayers (as shown by spontaneous desorption in Figure 5B), monitoring the transfer of unlabeled PA in at least one setting would strengthen the conclusion of the swap experiments.

Ups1-mediated transfer of PA has been demonstrated both by mass spectrometry analysis of donor and acceptor vesicles (Connerth et al., 2012) and by NBD-fluorescence-based lipid transfer assays (Lu et al., 2020; Miliara et al., 2015; Miliara et al., 2019; Miliara et al., 2023; Potting et al., 2013; Watanabe et al., 2015). The fluorescence-based approach has been the most widely applied across multiple studies and has enabled detailed analysis of various aspects of lipid transfer by Ups1. It has been used to investigate mutants of key structural elements—such as the lipid-binding pocket and the $\alpha 2$ -loop region. It has also been used to analyze fusion constructs between Ups1 and Mdm35, the influence of Mdm35 variants, and competition with excess Mdm35. Additionally, by comparing the transfer of NBD-labeled PA and NBD-labeled PS, this assay has provided insights into the determinants of the lipid specificity of Ups1. Hence, our experiments are based on the standard assay used to analyse lipid transfer in the field and thus can be correlated with the majority of published data.

Nevertheless, we agree that it is important to keep in mind that NBD labeling may alter the biophysical properties of lipids and, consequently, affect their transfer efficiency. Moreover, NBD-labeled lipids are not suitable for comparing the transfer efficiency of different PA species, as the label itself may mask differences in acyl chain composition. Therefore, it will be valuable to establish complementary methods that do not rely on NBD-labeled PA. We aim to develop these non-standard methods for possible inclusion in the present study, but even if not fully implemented at this stage, they will certainly form an important part of future investigations.

(4) The present study suggests that membrane domains with positive curvature at the outer membrane may serve as starting points for lipid transport by Ups1-Mdm35. Is anything known about the mechanisms that form such structures? This should be discussed in the text.

The origin of positively curved membrane domains is indeed highly relevant in the context of our findings, and while not the primary focus of this work, we will place more emphasis on discussing how such curvature may arise. Mechanisms include the action of curvature-generating proteins, asymmetric lipid composition and curvature induced at membrane contact sites. We have so far included examples of proteins in the outer mitochondrial membrane that are expected to influence curvature in their vicinity, and we will expand on this aspect and other contributing factors more thoroughly in the revised text.

Reviewer #2:

Summary:

Lipid transfer between membranes is essential for lipid biosynthesis across different organelle membranes. Ups1-Mdm35 is one of the best-characterized lipid transfer proteins, responsible for transferring phosphatidic acid (PA) between the mitochondrial outer membrane (OM) and inner membrane (IM), a process critical for cardiolipin (CL) synthesis in the IM. Upon dissociation from Mdm35, Ups1 binds to the intermembrane space (IMS) surface of the OM, extracts a PA molecule, re-associates with Mdm35, and moves through the aqueous IMS to deliver PA to the IM. Here, the authors analyzed the early steps of this PA transfer - membrane binding and PA extraction - using a combination of in vitro biochemical assays with lipid liposomes and purified Ups1-Mdm35 to measure liposome binding, lipid transfer between liposomes, and lipid extraction from liposomes. The authors found that membrane curvature, a previously

overlooked property of the membrane, significantly affects PA extraction but not PA insertion into liposomes. These findings were further supported by MD simulations.

Strengths:

The experiments are well-designed, and the data are logically interpreted. The present study provides an important basis for understanding the mechanism of lipid transfer between membranes.

Weaknesses:

The physiological relevance of membrane curvature in lipid extraction and transfer still remains open.

We thank the reviewer for the constructive feedback on our work. We agree that the physiological relevance of membrane curvature in lipid extraction and transfer remains an open question. Our data show that Ups1 binding to native-like OM membranes under physiological pH conditions is curvature-dependent, supporting the idea that this mechanism may optimize lipid transfer in vivo. While the intricate biophysical basis of this behaviour can only be dissected in vitro, these findings offer valuable insight into how curvature may functionally regulate Ups1 activity in the cellular context. To directly test this, it will be important in future studies to identify Ups1 mutants that lack curvature sensitivity and assess their performance in vivo, which will help clarify the physiological importance of this mechanism.

Reviewer #3:

The manuscript by Sadeqi et al. studies the interactions between the mitochondrial protein Ups1 and reconstituted membranes. The authors apply synthetic liposomal vesicles to investigate the role of pH, curvature, and charge on the binding of Ups1 to membranes and its ability to extract PA from them. The manuscript is well written and structured. With minor exceptions, the authors provide all relevant information (see minor points below) and reference the appropriate literature in their introduction. The underlying question of how the energy barrier for lipid extraction from membranes is overcome by Ups1 is interesting, and the data presented by the authors could offer a valuable new perspective on this process. It is also certainly a challenging in vitro reconstitution experiment, as the authors aim to disentangle individual membrane properties (e.g., curvature, charge, and packing density) to study protein adsorption and lipid transfer. I have one major suggestion and a few minor ones that the authors might want to consider to improve their manuscript and data interpretation:

Major Comments:

The experiments are performed with reconstituted vesicles, which are incubated with recombinant protein variants and quantitatively assessed in flotation and pelleting assays. According to the Materials and Methods section, the lipid concentration in these assays is kept constant at 5 μ M. However, the authors change the size of the vesicles to tune their curvature. Using the same lipid concentration but varying vesicle sizes results in different total vesicle concentrations. Moreover, larger vesicles (produced by freeze-thawing and extrusion) tend to form a higher proportion of multilamellar vesicles, thus also altering the total membrane area available for binding. Could these differences in the experimental system account for the variation in binding? To address this, the authors would need to perform the experiments either under saturation (excess protein) conditions or find an experimental approach to normalize for these differences.

We thank the reviewer for the constructive and positive comments. We agree that, since the total number of lipids was kept constant, the number of vesicles varied with vesicle size in our experiments. However, the setup was specifically designed to maintain a comparable total membrane surface area across conditions, ensuring a comparable number of available binding sites for Ups1. Because membrane surface area decreases with the square of the vesicle radius, keeping vesicle number constant would have led to a marked reduction in binding surface. Our approach was therefore aimed at preserving comparable binding capacity while varying membrane curvature.

With respect to multilamellarity, we thank the reviewer for addressing this important point. As described above, we aimed to maintain a constant total membrane surface area across all conditions to ensure an equal number of potential binding sites. We agree that multilamellarity in large liposomes could restrict accessibility to part of the membrane surface. However, we see in our experiments that even when the total membrane surface area of the small liposomes is reduced to one quarter of the standard amount, binding to the small liposomes remained stronger than to the larger liposomes at the higher concentration. This strongly indicates that restricted accessibility cannot account for the curvature-specific effect observed. Nonetheless, we will further address this aspect experimentally and in the discussion of the revised manuscript.

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